Can learned neighbor costs improve raw TP labels?

Compare raw six-channel HoVer-Net TP argmax with pixel WCSP. Keep model probabilities as unaries; learn only the neighbor-cost table from training GT.

The pixel experiment

26,896 pixels; all six labels at every pixel.

Aligned central 164 x 164 GT and predictions.

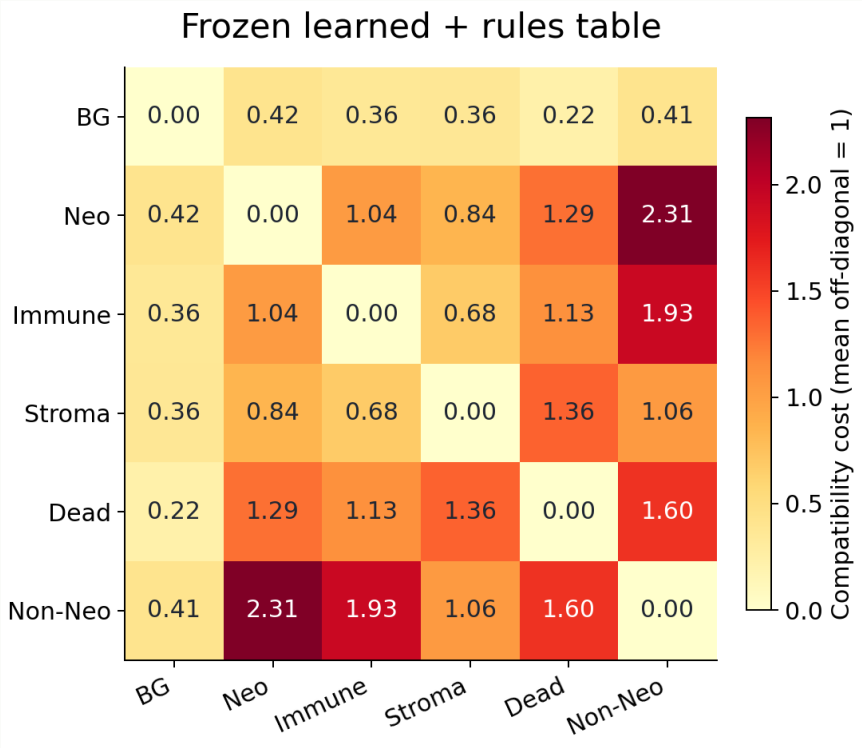
24 neighbors; nearer neighbors matter more.

Raster ICM; strict improvements; up to 100 rounds.

Train / test GT coverage

Class	Train pixels	Test pixels	Patches
Background	4,490,845	456,622	20/20
Neoplastic	426,677	45,575	10/20
Inflammatory	140,162	4,212	8/20
Connective	211,748	18,105	15/20
Dead	2,842	1,301	3/20
Non-Neoplastic	106,926	12,105	5/20

Tissues: 18 train; 19 test. No patch overlap.



Counts -> prevalence-adjusted incompatibility -> 80% data + 20% fixed rule prior -> normalize mean off-diagonal cost to 1.

Four soft preferences: same-type neighbors; isolate Dead from Non-Neoplastic; discourage Immune/Non-Neoplastic contact; favor connective tissue at tumor/healthy boundaries.

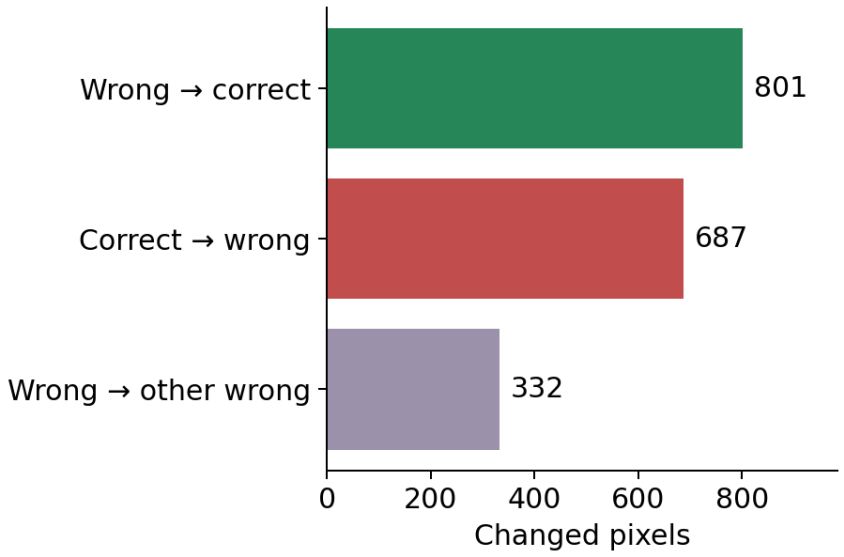
Held out from WCSP fitting only. HoVer-Net training-fold exposure and patient/slide independence could not be verified.

Results on the frozen 20-patch test set

Pooled metric	Raw TP	WCSP
Biological mIoU	0.3769	0.3752
Six-class mIoU	0.4731	0.4718
Pixel accuracy	0.9334	0.9336

Per-class IoU

Background	0.9545	0.9545
Neoplastic	0.6319	0.6338
Inflammatory	0.1604	0.1536
Connective	0.4130	0.4110
Dead	0.1554	0.1523
Non-Neoplastic	0.5237	0.5253



Changed pixels: 1,820 / 537,920

Biological mIoU change: -0.0017

Paired 95% CI: [-0.0050, +0.0080]

5,000 paired patch-bootstrap replicates

No reliable improvement: interval includes zero.

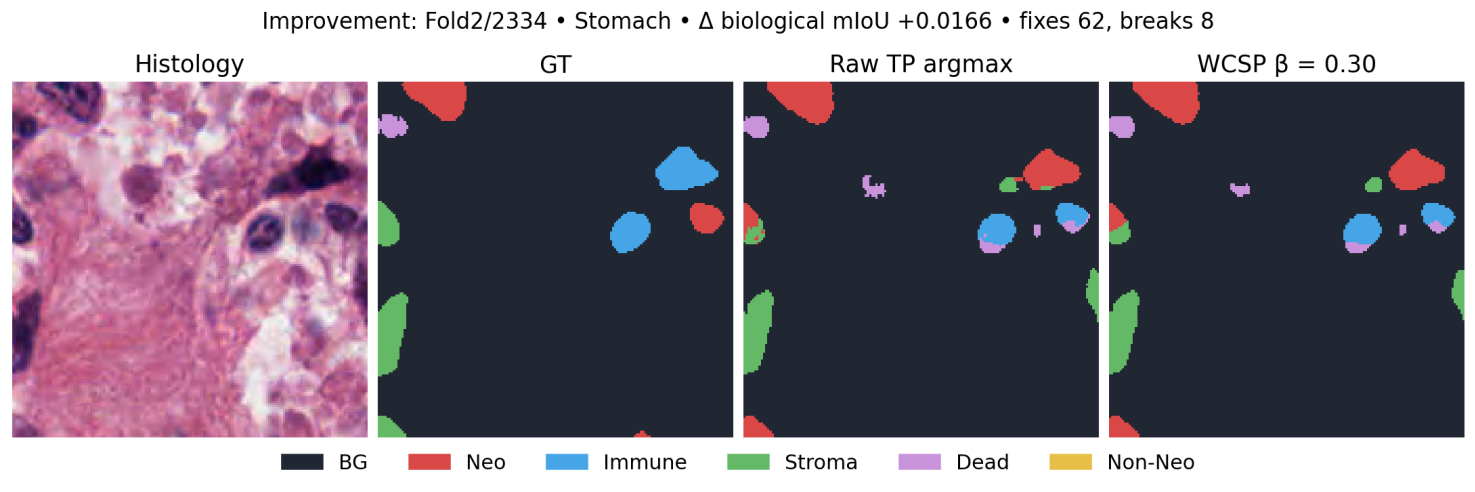
Background is 84.9% of GT pixels, so accuracy can hide biological-class errors. Biological mIoU gives each biological class equal weight. Pool confusion counts first; zero union gives IoU 0; no classes are dropped.

GT-absent classes: none. Rarest biological class: Dead, 1,301 pixels in 3 patches. Patch-bootstrap uncertainty does not account for unknown patient clustering.

Inference 17.0s; solver + audits 1.1s. Local minima 20/20; round limits 0.

Where the labels changed, and what this means

Examples are extremes by fixed five-class patch mIoU change; they do not represent every patch.



Limits: local pixel costs cannot enforce a stromal barrier or detect a dense immune cluster. ICM finds a single-pixel local minimum, not a global optimum. Coverage-enriched patches and unknown patient overlap limit generalization.

Office-hours questions: Are pixel contacts the right biological scale? How should rare Dead cells be protected? Would region-level context justify a separately preregistered experiment?